

JESUS F. SANCHEZ

jesusfsa@usc.edu | [jesusfsanchez.github.io](https://github.com/jesusfsanchez) | Los Angeles, CA | (805) 815-7654

EDUCATION

Master of Science - University of Southern California (Los Angeles, CA) Expected May 2027
Aerospace Engineering, Concentration: Propulsion (**GPA: 4.0**)
Bachelor of Science - California State University, Northridge (Northridge, CA) August 2024
Mechanical Engineering, Concentration: Aerospace (**GPA: 3.7**)

WORK EXPERIENCE

NASA Jet Propulsion Laboratory, *Mechanical Engineering Intern* May 2022 - February 2024

- Led the design of a 3U electromechanical systems from prototype to engineering model for the Mars' Sample Retrieval Lander (SRL), focusing on design for manufacturability and assembly (DFM & DFA)
- Generated CAD models, performed tolerance stack-up analysis, created dozens of engineering drawings using GD&T in Siemens NX and authored build, assembly, and test instructions
- Collaborated closely with the manufacturing team to support fabrication and troubleshoot hardware assembly issues, resulting in accurate and timely delivery of hardware

ENGINEERING PROJECTS & RESEARCH

Liquid Propulsion Laboratory (LPL), *Propulsion Engineer* September 2024 - Present

- Designed and tested a regeneratively cooled 3 kN Jet-A/LOX rocket engine focusing on thrust chamber geometry, nozzle contouring, and injector design; achieved 95% c^* efficiency in successful hotfire testing
- Developed a bolt analysis tool to determine optimal bolt count, size, and torque to maintain a sealed chamber-injector interface under operational loads
- Conducted FEA of the thrust chamber assembly in Ansys Mechanical to evaluate stresses and seal integrity, leading to design revisions that minimized interface gap formation
- Performed CFD simulations to optimize injector geometry by reducing pressure losses and improving flow distribution
- Built MATLAB-based thermal models to estimate maximum allowable firing duration and back-calculate combustion gas temperatures from thermocouple testing data
- Automated hotfire data analysis with custom MATLAB scripts to evaluate performance metrics, thermal margins, and guide design updates, including optimized mixture ratio selection based on empirical results

Novel Scalable Swirl Hydrogen Combustor, *Research Fellow* May 2023 - August 2024

- Researched and developed a novel swirl-injection combustor for gas turbine engines to improve air-fuel mixing and achieve stable hydrogen combustion
- Designed combustor geometry in SOLIDWORKS and performed CFD simulations in Ansys Fluent to predict swirl length and flame behavior
- Experimentally evaluated combustor performance achieving up to 60% hydrogen dilution, exceeding current industry findings by 10%

ASME Human Powered Vehicle (HPV) Competition, *Fairing Team Member* August 2023 - May 2024

- Designed, analyzed (CFD), and fabricated a low-drag carbon fiber fairing with a drag coefficient of 0.43, a performance improvement over previous year's 0.50
- Performed FEA to evaluate and optimize fairing mounting arms, resulting in a 13% weight reduction and reduced stress concentrations

NASA MINDS Competition, *Mechanical Design Sub-Team Member* September 2020 - May 2022

- **2022 CubeSat**: Conducted finite element analysis through SOLIDWORKS to optimize the design of a 4-inch cube satellite structure for weight reduction and material selection
- **2021 Autonomous Robot**: Designed the drivetrain system and analyzed (FEA) the chassis of a robot intended for nondestructive testing of space vehicles, enabling it to traverse 90-degree surfaces, achieving 3rd place in competition

SKILLS

Software: CAD (Siemens NX, SOLIDWORKS), CFD (Ansys Fluent, OpenFOAM), FEA (Ansys Mechanical, SOLIDWORKS Simulation), Cantera, LabVIEW, MATLAB, Python, Simulink, Teamcenter

Technical: Geometric Dimensioning and Tolerancing (GD&T), Tolerance Analysis, 3D Printing

COURSEWORK

Spacecraft Design, Compressible Gas Dynamics, Intro to Computational Fluid Dynamics, Fundamentals and Applications of Combustion, Partial Differential Equations, Rocket Propulsion, Aeropropulsion (see [more](#))